

Lecture 10: The Chain Rule

GOAL: Be able to differentiate functions of the form $f(g(x))$.

Summary:

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

$$\frac{d}{dx}[f(g(x))] = \frac{df}{dg} \cdot \frac{dg}{dx}$$

$$[g(x)^p]' = p[g(x)]^{p-1} \cdot g'(x) \quad [b^x]' = b^x \cdot \ln b$$

$$[\sin(\theta^\circ)]' = \frac{\pi}{180} \cos(\theta^\circ)$$

